

Secure Face Registration and Verification Framework

using Face Quality Assessment and Liveness Detection

Final Project Report -- Nishika

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1. The Problem

Think of this project as building the brain of a security guard for face-based login. The guard has two jobs. First, when someone signs up (registration), take a clear photo of their face and remember it. Second, when someone tries to log in later (verification), look at their face again and answer two questions: is this a real, live person in front of the camera right now, and does this face actually belong to the person who signed up.

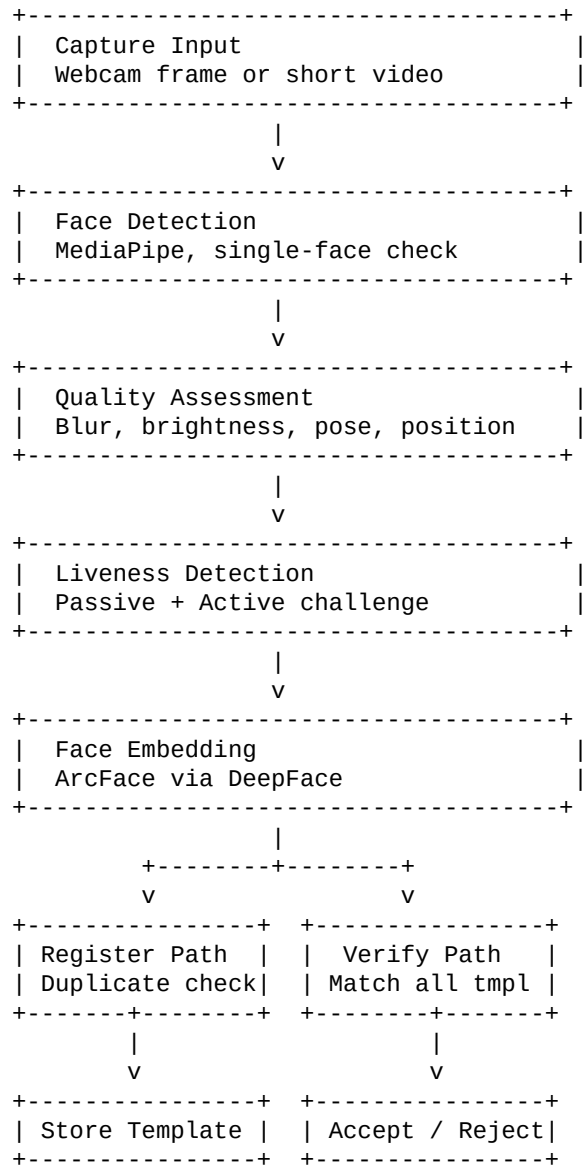
The reason this is hard is that a face-matching system alone can be tricked. Someone could hold up a printed photo, play a video, or display a photo on a phone screen, and a system that only checks whether a face matches might be fooled. So the system also needs a way to catch these tricks, called liveness detection, and a way to make sure the photo it is working with is actually usable in the first place, called image quality assessment.

The project brief splits this into two phases, verification and registration. This design merges them into one shared pipeline with two entry points, since both phases reuse the same quality, liveness, and matching logic. The full pipeline is shown in Figure 1, Section 2.

2. High-Level Architecture

Both registration and verification share every stage up to Face Embedding, then split into two separate paths, shown in Figure 1 below.

Figure 1 -- Shared Core Pipeline



3. Face Matching and Liveness Detection - Measured Results

Table 1: Face Matching Accuracy (Frontal-vs-Frontal, the production verification path)

Metric	Value
Equal Error Rate (EER)	3.19% at threshold 0.2642
ROC AUC	0.9953
False Accept Rate (at threshold 0.40)	0.34% (2/595)
False Reject Rate (at threshold 0.40)	15.09% (16/106)
Half Total Error Rate (HTER)	7.72%

Table 2: Passive Liveness / Attack Detection (n=76 attacks, n=75 genuine)

Metric	Value
APCER (attacks wrongly accepted, at 0.90)	46.05% (35/76)
BPCER (genuine wrongly rejected)	0.00% (0/75)
ACER (average classification error)	0.230

These numbers were re-measured on an expanded sample this pass, superseding an earlier n=5 figure that was too small to trust. The n=76/n=75 result came out worse than the earlier estimate and is reported as measured, not adjusted to look better.

Per-attack-type breakdown appears in Table 3, Section 4, which shows the aggregate ACER number hides a very uneven picture across attack types.

Figure 1 caption: both paths share every stage up to Face Embedding, then split -- register (store a new template) or verify (match against stored templates, accept or reject). Colors in the source diagram (docs/architecture_diagram.png): gray = capture/detection, teal = quality/liveness, purple = embedding, orange = register, pink = verify.

4. Attack Scenarios and Testing

This project addresses physical presentation attacks (print, replay, video) only. Digital attacks such as deepfake injection are a distinct category and are explicitly out of scope.

Table 3: Attack Scenario Results

Attack Type	Result	Notes
Printed Photo	Caught (5.3% missed)	Reliably caught
Screen Replay (Phone)	Caught (5.3% missed)	Reliably caught
Screen Replay (Laptop)	73.7% missed	REAL, CONFIRMED WEAKNESS
Frozen Frame	100% missed	Expected -- active liveness's job
Recorded Gesture	Bypasses active check	Landmark-based, known limit
Multiple Face	Rejected	Face detection stage
Poor Quality	Rejected	Quality assessment stage

A threshold sweep shows raising the deployed liveness threshold to 0.95-0.96 is a free improvement in this sample (genuine users unaffected), but no threshold fully closes the laptop-replay gap without a real cost to genuine acceptance. This report does not change the deployed threshold -- that decision is left to the project owner with this evidence in hand.

5. Limitations and Scope Decisions - Stated Plainly

Physiological liveness (rPPG) - deferred. Built and calibrated offline but not wired into the live pipeline; it needs a stable 10-15 second capture window that does not fit the current per-tick UI model.

Active liveness covers 2 challenge types (blink, head-turn). A direct test confirmed a replayed video of a real blink is accepted -- landmark-based active liveness cannot distinguish a live blink from a recorded one, exactly why the system does not rely on it alone.

Occlusion detection is a visibility-based approximation, not a trained classifier. It can flag likely obstruction but not identify what is causing it.

Demographic bias - confirmed, unmitigated. Measured across 80 real images: quality pass rate 20.0% (female) vs. 42.5% (male); liveness pass rate 25.0% (senior) vs. 46.2% (middle-aged); a smaller but present skin-tone gap. No mitigation attempted.

Multi-angle enrollment simplified to front-only. The original three-angle flow was unreliable in practice, and duplicate-detection and verification already relied on the front template almost exclusively for accuracy.

A known, intermittent WebRTC connection hiccup exists -- the camera connection can silently drop for 20-30 seconds and self-recover. Root-caused as far as the evidence allows to a library-level characteristic, not an application bug.

6. Technology Stack

Language: Python 3.11. Computer Vision: OpenCV. Face Detection and Landmarks: MediaPipe Face Mesh, 468-point landmarks. Face Recognition and Matching: DeepFace with the ArcFace backend, pretrained, no retraining. Passive Liveness: DeepFace anti-spoofing (MiniFASNet). Active Liveness: custom MediaPipe + EAR + solvePnP. Backend API: FastAPI (/register, /verify). Database: SQLite, local, no cloud dependency. User Interface: Streamlit + streamlit-webrtc, chosen over React since the brief marks the UI as optional.

7. Verification: Fresh Automated Test Suite

Full suite run fresh for this report:

13 passed in 10.10s

Covering encryption round-trip integrity, consent enforcement, soft/hard delete behavior, embedding similarity math, template matching, and quality/liveness structure on genuine captures.

8. Learning Outcomes

Computer Vision, Face Recognition, Liveness Detection, Biometric Authentication, API Development, and Performance Evaluation -- each grounded in this project's own measured results rather than generic claims, including the discipline of re-measuring a suspiciously small sample and reporting the result even when it came out worse than expected.

This report consolidates findings already documented in README.md, Evaluation_Report.md, and scope_decision_worksheet.md.